

Instructions.

1. **Due April 30, 11:59 PM uploaded to Blackboard as a single pdf and a single ZIP archive. This deadline is firm. Late uploads will not be accepted.**
2. This project will be done in a team of two members. You may **NOT** discuss or consult with any students outside your team.
3. All work must be typeset in L^AT_EX. Handwritten or scanned submissions will not be accepted.
4. All figures must have axis labels and descriptive captions. Tables, if any, must have descriptive captions.
5. This document may be updated if a mistake is found or further clarification is needed. A Blackboard announcement will be posted as soon as this document is updated.
6. Submission guidelines.
 - (a) Label your report as ENME408_S26_Project03_<lastnames>.pdf.
 - (b) Include all figures and tables with captions in your report.
 - (c) You may include MATLAB scripts and representative outputs as Appendices. Label the appendices and cite them in your report appropriately.
 - (d) Place all TEX files, MATLAB scripts, and Simulink models in a folder, then archive it. Label your archive as ENME408_S26_Project03_<lastnames>.zip.
 - (e) Upload your pdf file and the archive folder to Blackboard before the deadline.
7. Honor code: “I have neither received nor given aid to anyone outside my team during this project.”

Sign/print your names here.

If you cannot sign/print in the box above, please write the honor code on the first page of your submission and sign it.

8. Reports with substantial similarity will be reported to the UMBC Academic Conduct Committee.
9. Include a **credit author statement** in your report. The statement recognizes individual author contributions and facilitates collaboration. For example,
Author 1: Methodology, Software, Visualization
Author 2: Data curation, Writing, Original draft preparation, Validation.

Hilbert-Trajectory Generation and Tracking

In this step, you will use the cascaded guidance and control architecture developed above to generate and track a planar trajectory based on a second-order Hilbert curve. The objective is to construct a nontrivial reference trajectory with repeated changes in direction, and then evaluate the ability of the closed-loop multicopter system to track that trajectory.

1. **Waypoint generation using a second-order Hilbert curve.** Construct a second-order Hilbert curve in the horizontal plane and use its vertices as a sequence of waypoints for the translational position reference. Denote the reference position by

$$p_{\text{ref}}(t) = \begin{bmatrix} p_{1,\text{ref}}(t) \\ p_{2,\text{ref}}(t) \\ p_{3,\text{ref}}(t) \end{bmatrix}. \quad (1)$$

Let $p_{1,\text{ref}}$ and $p_{2,\text{ref}}$ follow a second-order Hilbert curve with branch length equal to 2 m. Thus, consecutive horizontal waypoints must differ by 2 m along either the p_1 or p_2 direction. Let the vertical component vary during the maneuver so that

- during the first half of the trajectory, $p_{3,\text{ref}}$ increases by 1 m, and
- during the second half of the trajectory, $p_{3,\text{ref}}$ decreases by 1 m.

Furthermore, consider a yaw reference $\psi_{\text{ref}}(t)$ which aligns the vehicle heading with the next waypoint. Denote the waypoint sequence by

$$p_i = \begin{bmatrix} p_{1,i} \\ p_{2,i} \\ p_{3,i} \end{bmatrix}, \quad \psi_i, \quad i = 1, \dots, N, \quad (2)$$

where $N = 16$ is the total number of waypoints in the 2nd-order Hilbert trajectory. Use the waypoint sequence given in Table 1. The horizontal coordinates define a second-order Hilbert curve with branch length 2 m, while the vertical coordinate introduces the required climb-and-descent profile. The yaw waypoints align the vehicle heading with the next position waypoint.

Table 1: Waypoint sequence for the modified second-order Hilbert trajectory.

Waypoint i	$p_{1,i}$ [m]	$p_{2,i}$ [m]	$p_{3,i}$ [m]	ψ_i [deg]
1	0	0	0.00	90
2	0	2	-0.25	90
3	2	2	-0.50	0
4	2	0	-0.75	-90
5	4	0	-1.00	0
6	6	0	-1.00	0
7	6	2	-1.00	90
8	4	2	-1.00	180
9	4	4	-1.00	90
10	6	4	-0.75	0
11	6	6	-0.50	90
12	4	6	-0.25	180
13	2	6	0.00	180
14	2	4	0.00	270
15	0	4	0.00	180
16	0	6	0.00	90

Plot the waypoint sequence in the (p_1, p_2) plane and clearly indicate the order in which the waypoints are traversed. In addition, provide a 3D plot of the waypoint sequence to illustrate the combined horizontal Hilbert pattern and vertical motion.

2. **Guidance law with velocity and acceleration constraints.** Generate a smooth reference trajectory between successive waypoints using a guidance law with bounded velocity and bounded acceleration as shown in Appendix A of [1] (also seen in class). Set the maximum velocity constraint $v_{p,\max} = 1$ m/s and the maximum acceleration constraint $a_{p,\max} = 1$ m/s². Furthermore, set maximum yaw rate constraint $v_{\psi,\max} = 2$ rad/s and the maximum yaw acceleration constraint $a_{\psi,\max} = 0.5$ rad/s². Consider that the waypoints are reached when the multicopter enters a radius of $R_p = 0.15$ m around the current waypoint. This guidance law produces a trajectory consisting of acceleration, cruise, and deceleration phases.
3. **Closed-loop simulation of the full nonlinear multicopter dynamics.** Use the cascaded controller developed in the previous steps to track the Hilbert trajectory. In particular, simulate the full nonlinear multicopter dynamics from the previous project, where the outer loop generates the virtual force command, the virtual force is mapped to thrust magnitude and attitude references, and the inner loop generates the torque command. Use the guidance law from the previous part to generate the reference position, velocity, and acceleration along the Hilbert trajectory.
4. **Required plots.** Include the following plots in your report.
 - (a) The reference and actual trajectories in the horizontal plane (2D trajectory).
 - (b) A 3D trajectory plot using `plot3`. Use `quiver3` to plot the velocity vectors and the yaw, as shown in class.
 - (c) The position components $p_1(t), p_2(t), p_3(t)$ with respect to t together with their reference signals (3 subplots).
 - (d) The tracking errors

$$e_p(t) \triangleq p(t) - p_{\text{ref}}(t). \quad (3)$$

- (e) The roll and pitch angles together with their commanded reference angles, that is, $\phi(t)$ and $\phi_{\text{ref}}(t)$ on one plot, and $\theta(t)$ and $\theta_{\text{ref}}(t)$ on another plot.

All figures must have labeled axes and descriptive captions.

5. **Tracking performance and discussion.** Briefly discuss the closed-loop tracking performance. In particular, comment on:
 - (a) where the largest tracking errors occur along the trajectory,
 - (b) the effect of sharp turns and repeated direction changes,
 - (c) whether the inner-loop attitude controller is able to track the commanded roll and pitch references sufficiently well, and
 - (d) limitations of the cascaded control design observed in this maneuver.

Remark 4. A second-order Hilbert curve provides a compact trajectory with repeated turns and motion in both horizontal directions. It therefore serves as a useful benchmark for evaluating the guidance, position-control, and attitude-control layers together in a single closed-loop simulation.

References

- [1] J. Spencer, J. Lee, J. A. Paredes, A. Goel, and D. Bernstein, “An adaptive pid autotuner for multicopters with experimental results,” in *2022 International Conference on Robotics and Automation (ICRA)*, IEEE, 2022, pp. 7846–7853.